Heterodyne Detection of Axion Dark Matter

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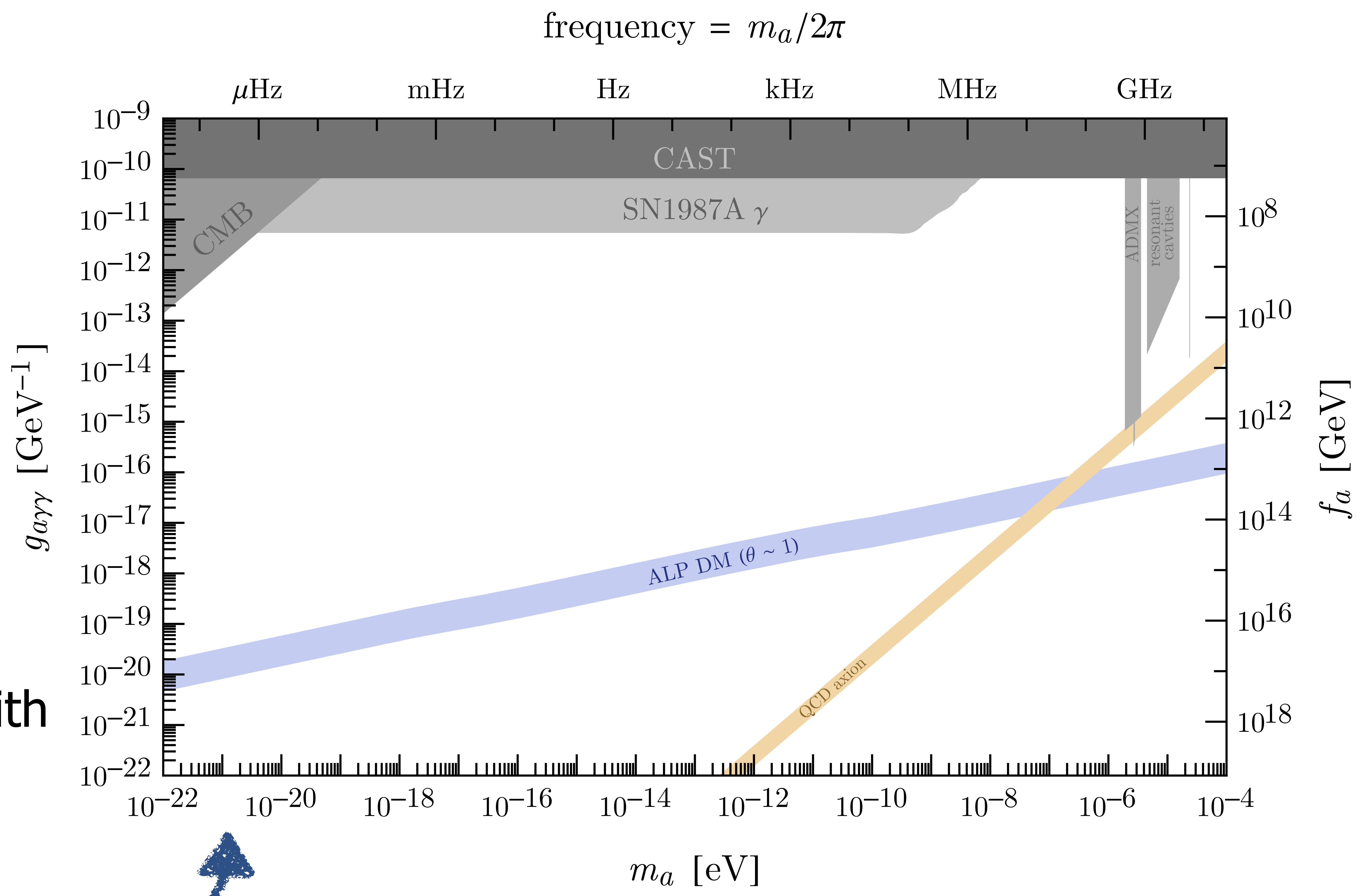
Axion Dark Matter

For this talk: axions are light pseudoscalar fields protected by a shift symmetry

Theoretically well-motivated, with simple and predictive dark matter production mechanisms

Especially interesting regions of parameter space:

- QCD axion (solves strong CP problem)
- ALP DM (produced in appropriate abundance with $\mathcal{O}(1)$ misalignment angle, T-independent mass)
- Fuzzy DM (helps resolve galactic-scale astrophysical issues)



Electromagnetic Axion Searches

Axions generically have electromagnetic coupling

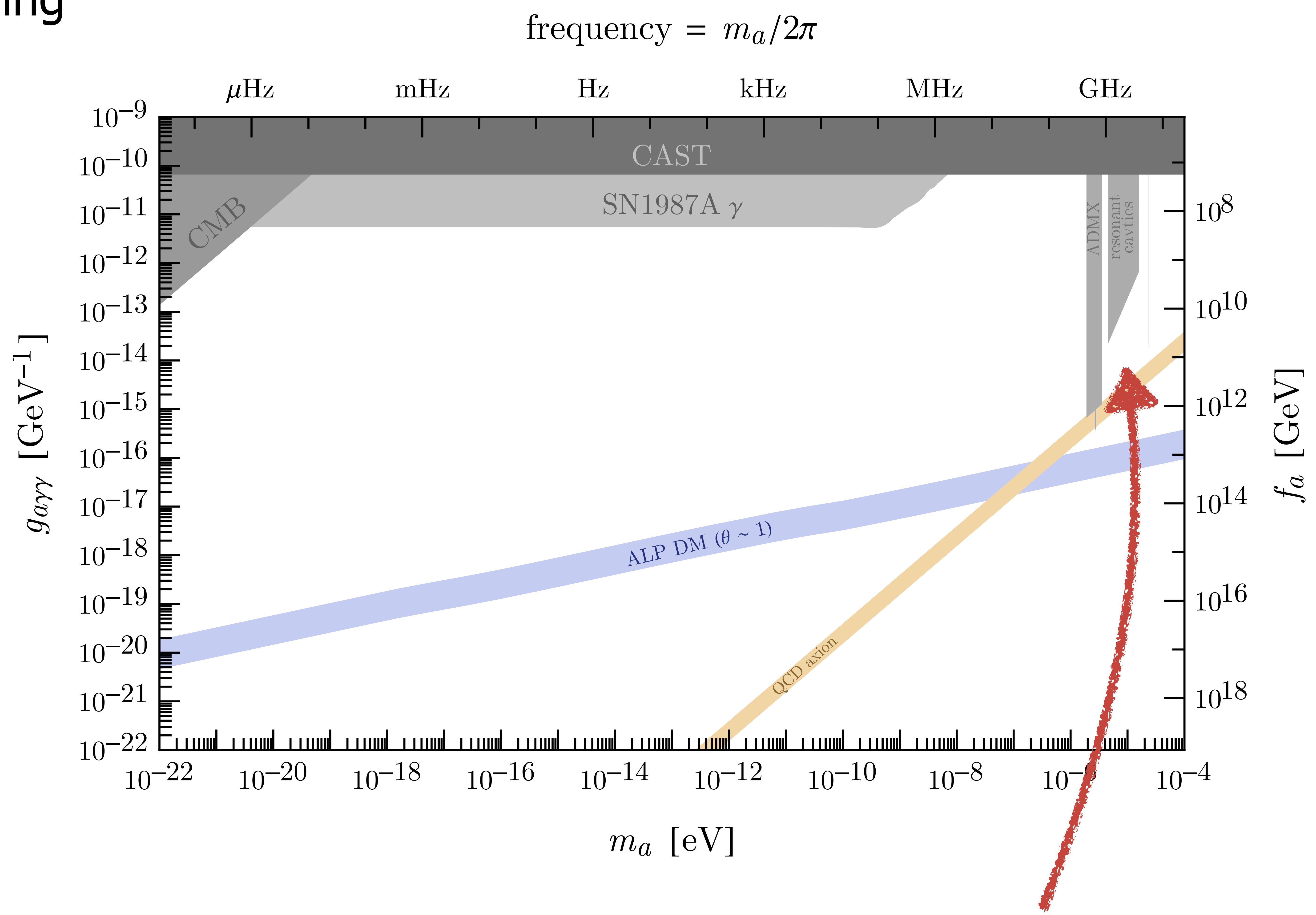
$$\mathscr{L} \supset g_{a\gamma\gamma}\, aF\tilde{F} \sim \mathbf{J}_{\text{eff}} \cdot \mathbf{A}$$

which produces an effective current

$$\mathbf{J}_{\text{eff}} = g_{a\gamma\gamma}\, \dot{a}\mathbf{B}$$

For static $\mathbf{B}$, current oscillates at

$$\omega \sim (1\ \text{GHz}) \frac{m_a}{\mu\text{eV}}$$



Sikivie haloscope: resonantly amplify $\mathbf{J}_{\text{eff}}$ in microwave cavity

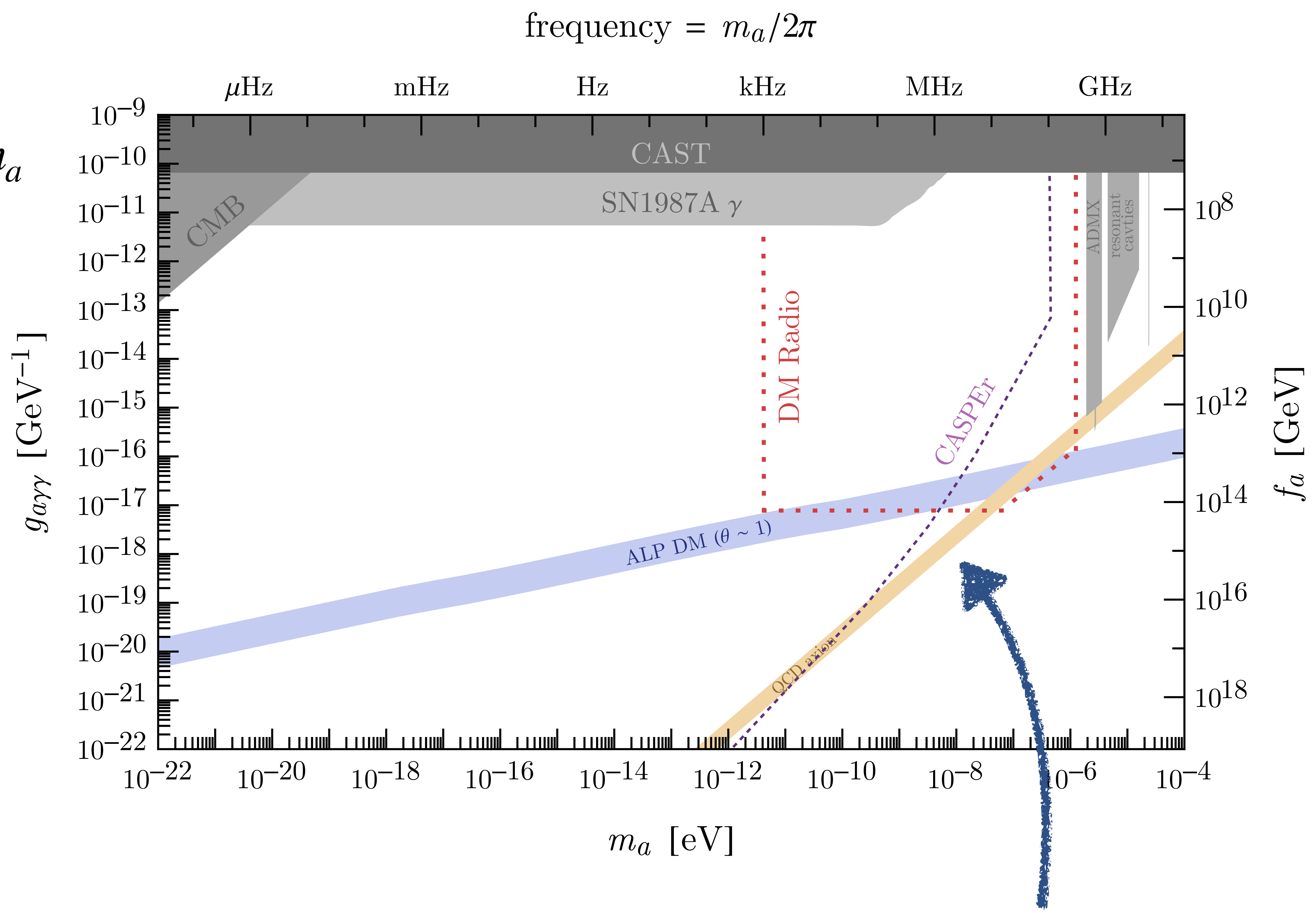
Challenges at Low Masses

At low mass: cavity would need huge size $L \sim 1/m_a$

Must decouple resonant frequency from size

DM Radio/ABRA: use LC circuit with resonant frequency $\omega \ll 1/L$

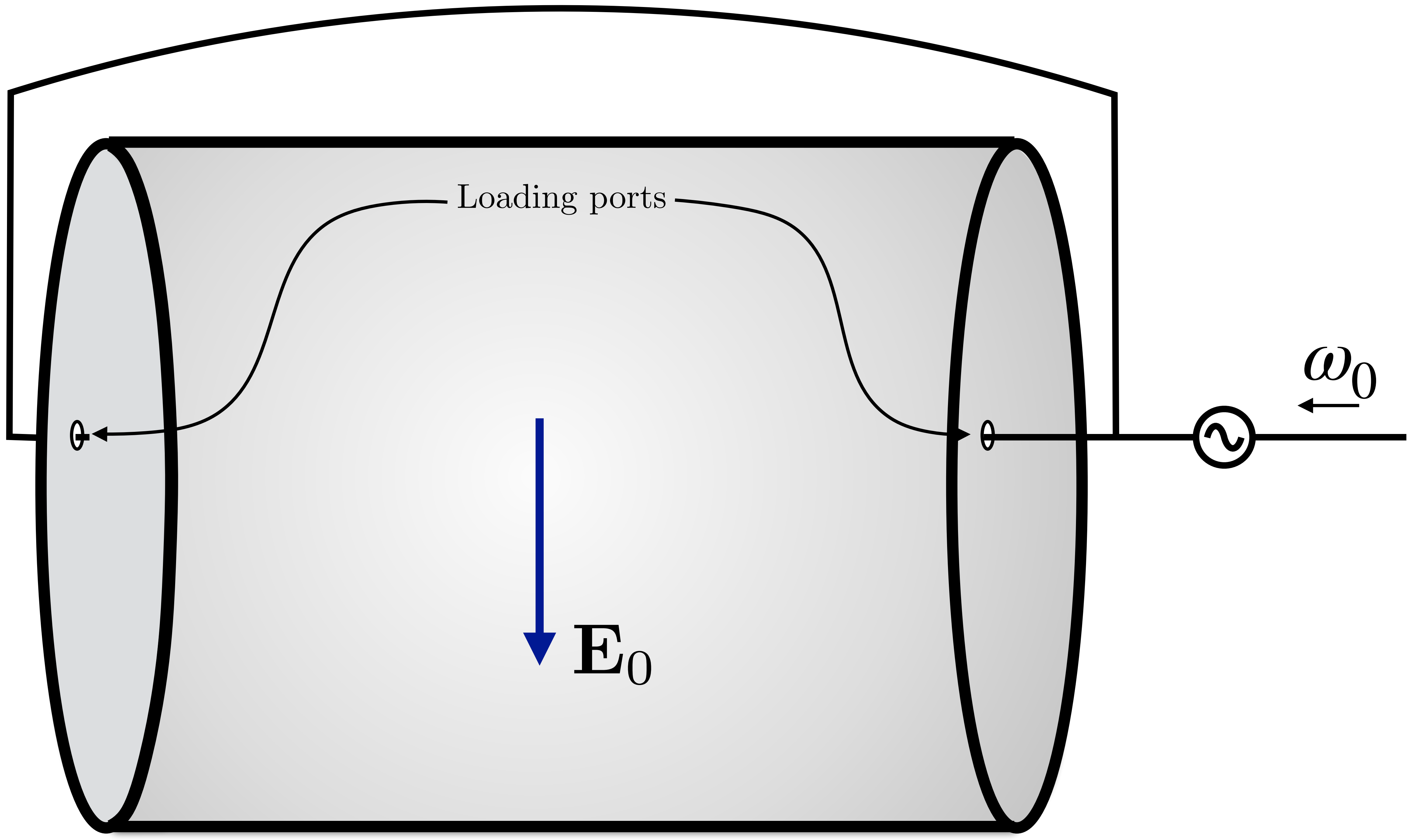
CASPEr: use NMR setup (QCD axion only)



Leaves gap in QCD axion coverage, limited ALP DM coverage

Resonant Heterodyne Detection

Load a cavity mode at frequency $\omega_0 \sim \mathrm{GHz} \gg m_a$



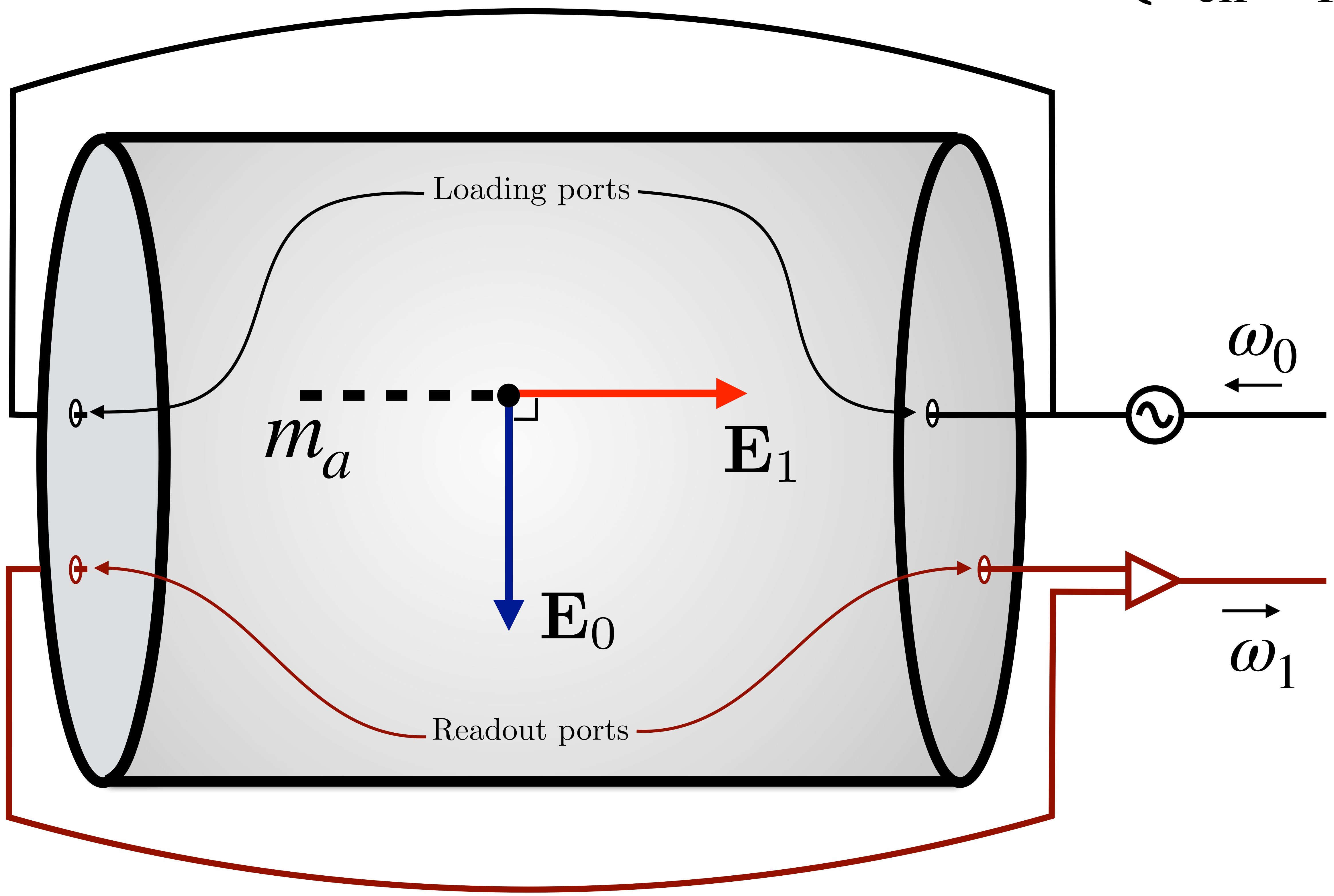
Resonant Heterodyne Detection

Induced axion effective current $\mathbf{J}_{\text{eff}} = g_{a\gamma\gamma}\,\dot{a}\mathbf{B}$ oscillates at frequency $\omega_1 = \omega_0 + m_a$

Resonantly drives power into an orthogonal "signal" mode with frequency ω_1

Axion-induced emf is parametrically larger!

$$\mathscr{E}_a \sim \dot{B}_a \sim \begin{cases} J_{\text{eff}}\, m_a & \text{static field LC} \\ J_{\text{eff}}\, \omega_1 & \text{heterodyne} \end{cases}$$



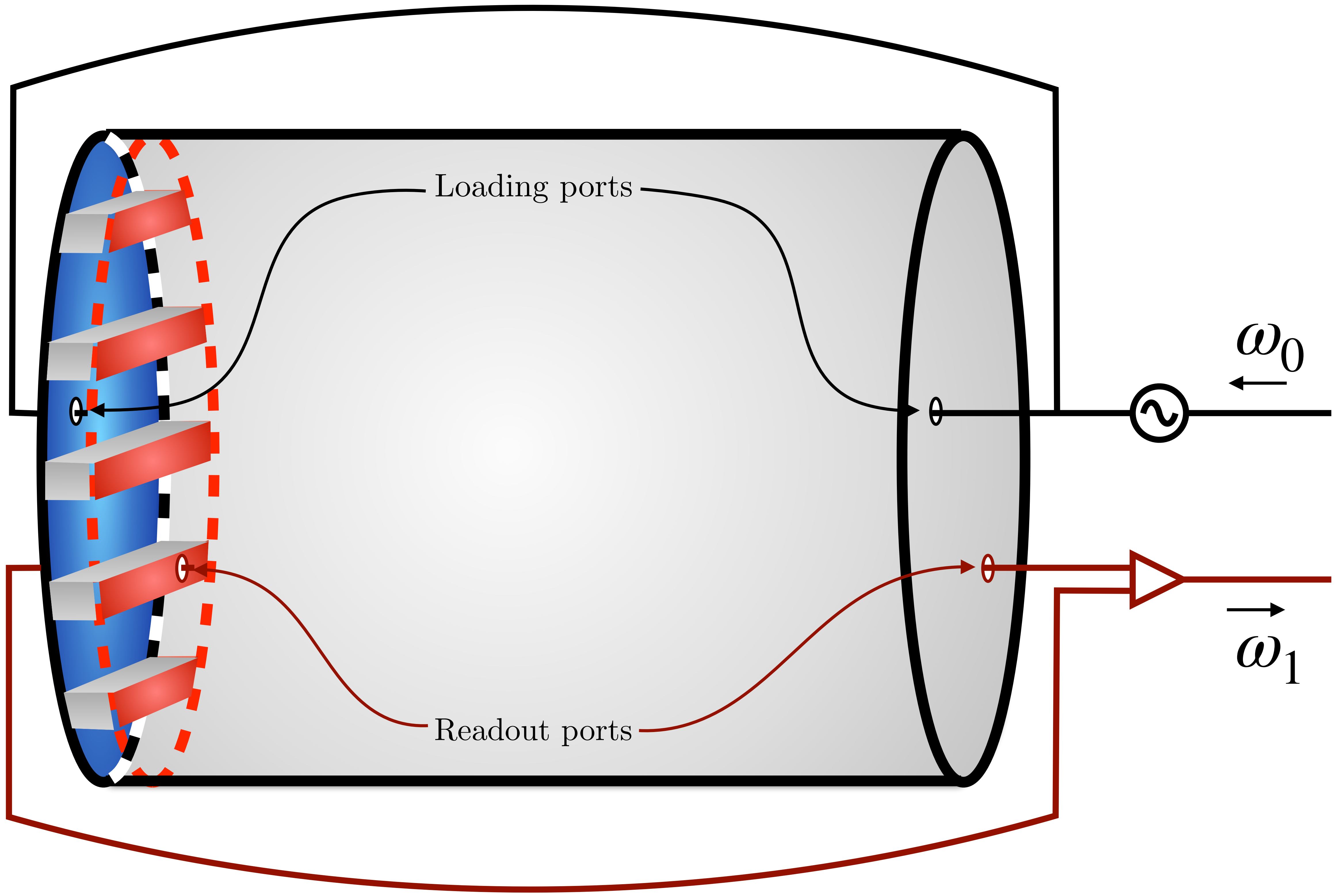
1912.11048: A. Berlin, R. Tito D'Agnolo, S. Ellis, C. Nantista, J. Neilson, P. Schuster, S. Tantawi, N. Toro, K. Zhou

see also: 1912.11056

Resonant Heterodyne Detection

Scan in m_a by perturbing cavity walls

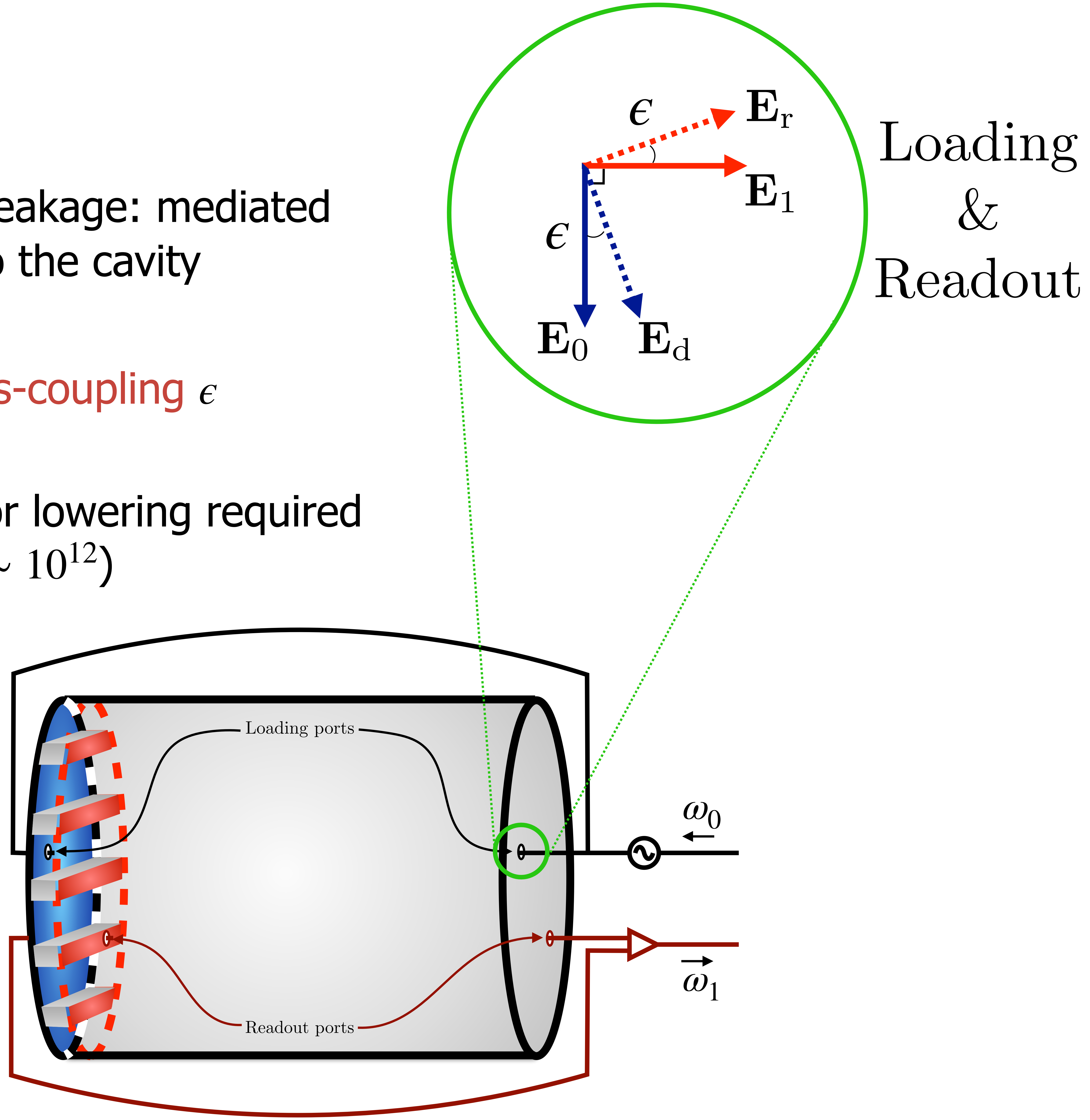
Many orders of magnitude in m_a scanned by small fractional changes in ω_0, ω_1



Main noise source at low m_a is leakage: mediated by the power loaded into the cavity

Suppressed by small cross-coupling ϵ

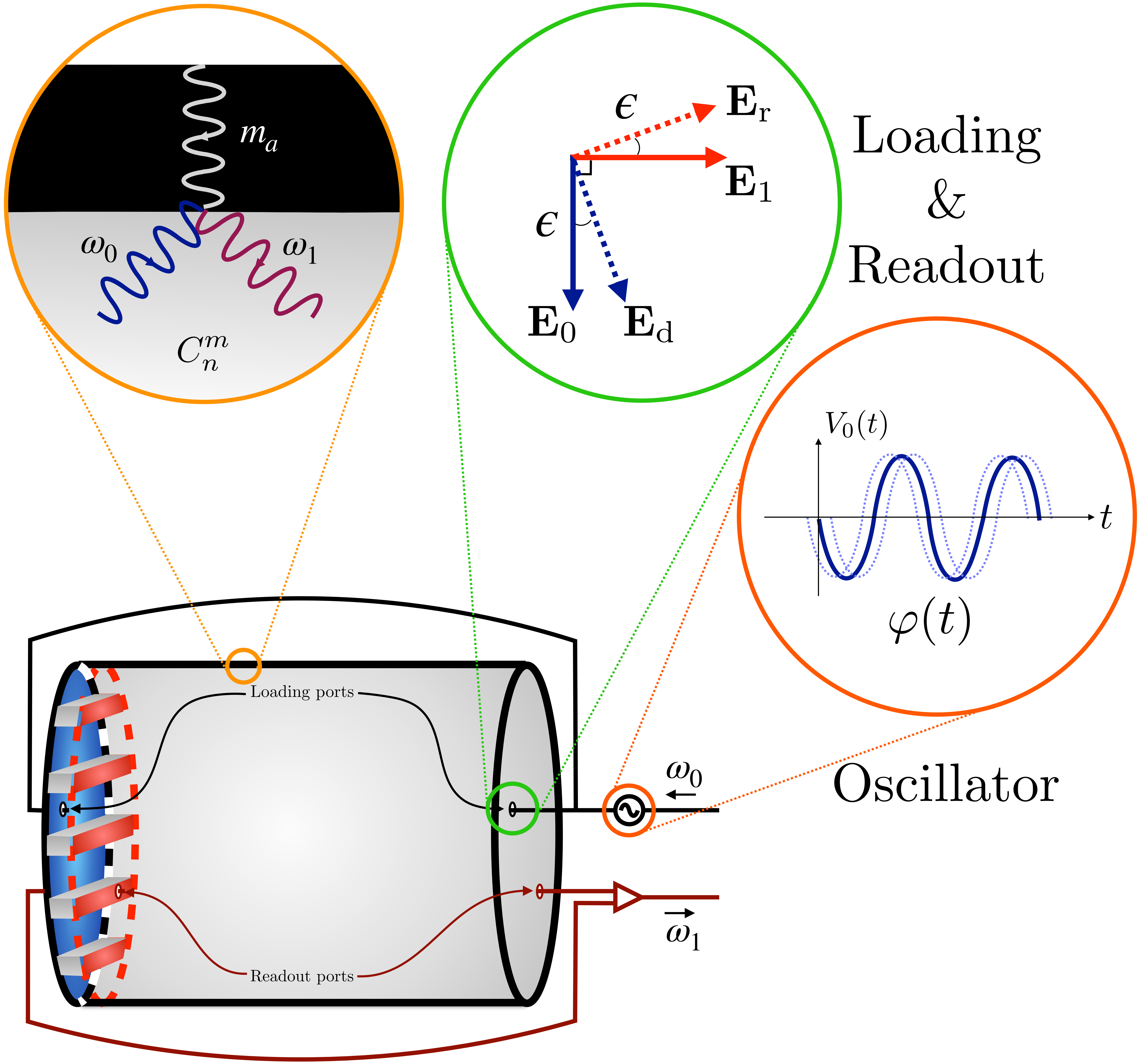
Motivates use of SRF cavity for lowering required input power ($Q \sim 10^{12}$)

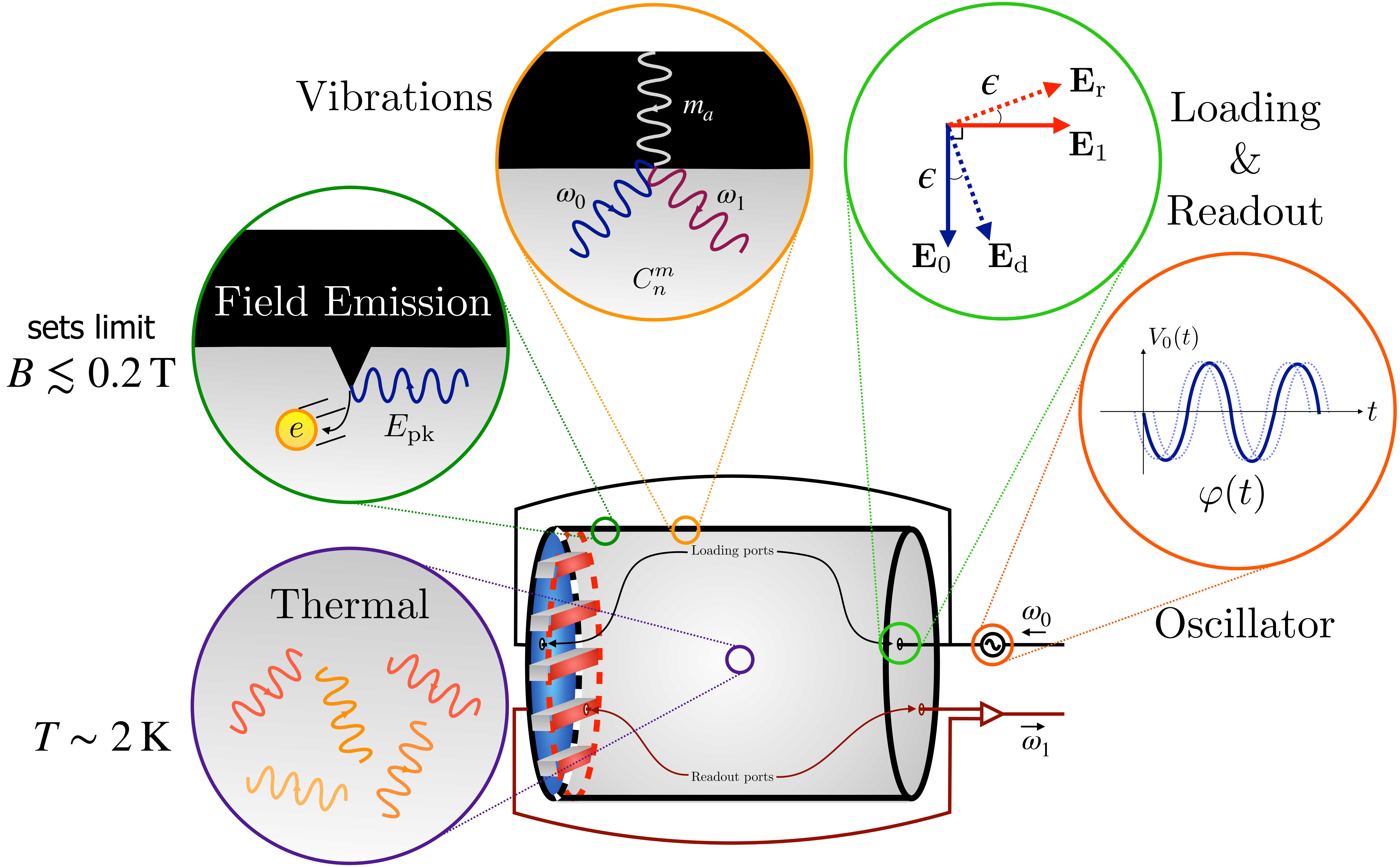


Direct environmental noise at ω_1 negligible

Leakage arises from upconversion of noise at frequency m_a to ω_1, like the axion signal

Dominates for low m_a




Vibrations
m_a
ω_0
ω_1
C_n^m
ϵ
$\mathbf{E}_\mathrm{r}$
$\mathbf{E}_1$
ϵ
$\mathbf{E}_0$
$\mathbf{E}_\mathrm{d}$
Loading & Readout
sets limit
$B \lesssim 0.2\,\mathrm{T}$
Field Emission
e
E_pk
$V_0(t)$
t
$\varphi(t)$
Oscillator
Loading ports
ω_0
Thermal
$T \sim 2\,\mathrm{K}$
$\vec{\omega}_1$
Readout ports

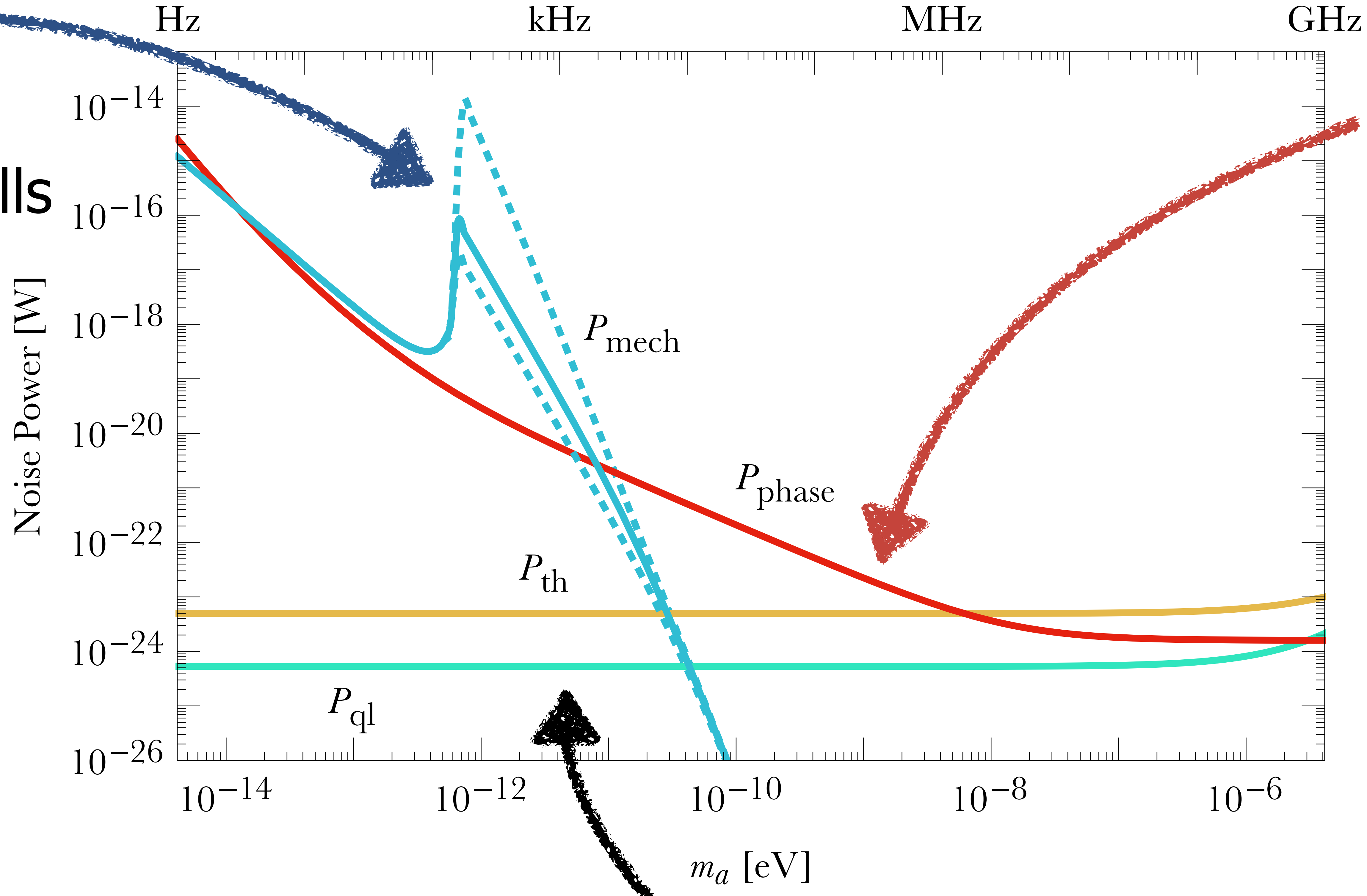
Noise Estimates

Total noise power for a single scan step probing axions at mass m_a

Mechanical vibrations of walls at frequency m_a

$$P_{\text{mech}} \sim \epsilon^2 \delta^2 \frac{\omega_0^2 \omega_{\text{min}}^3}{m_a^5} P_{\text{in}}$$

($\epsilon = 10^{-5}$ shown)



Phase noise in loading oscillator at frequency m_a

$$P_{\text{phase}} \sim \epsilon^2 S_\varphi(m_a) \frac{\omega_0}{Q} P_{\text{in}}$$

Usual thermal and quantum noise, dominates for high m_a

Experimental Precedent

MAGO (2005): heterodyne gravitational wave detection, reported $\epsilon \sim 10^{-7}$

DarkSRF (2020): dark photon light-shining-through-wall experiment with SRF cavities, stabilized to $\Delta\omega/\omega \sim \delta \sim 10^{-10}$

Resonant Reach

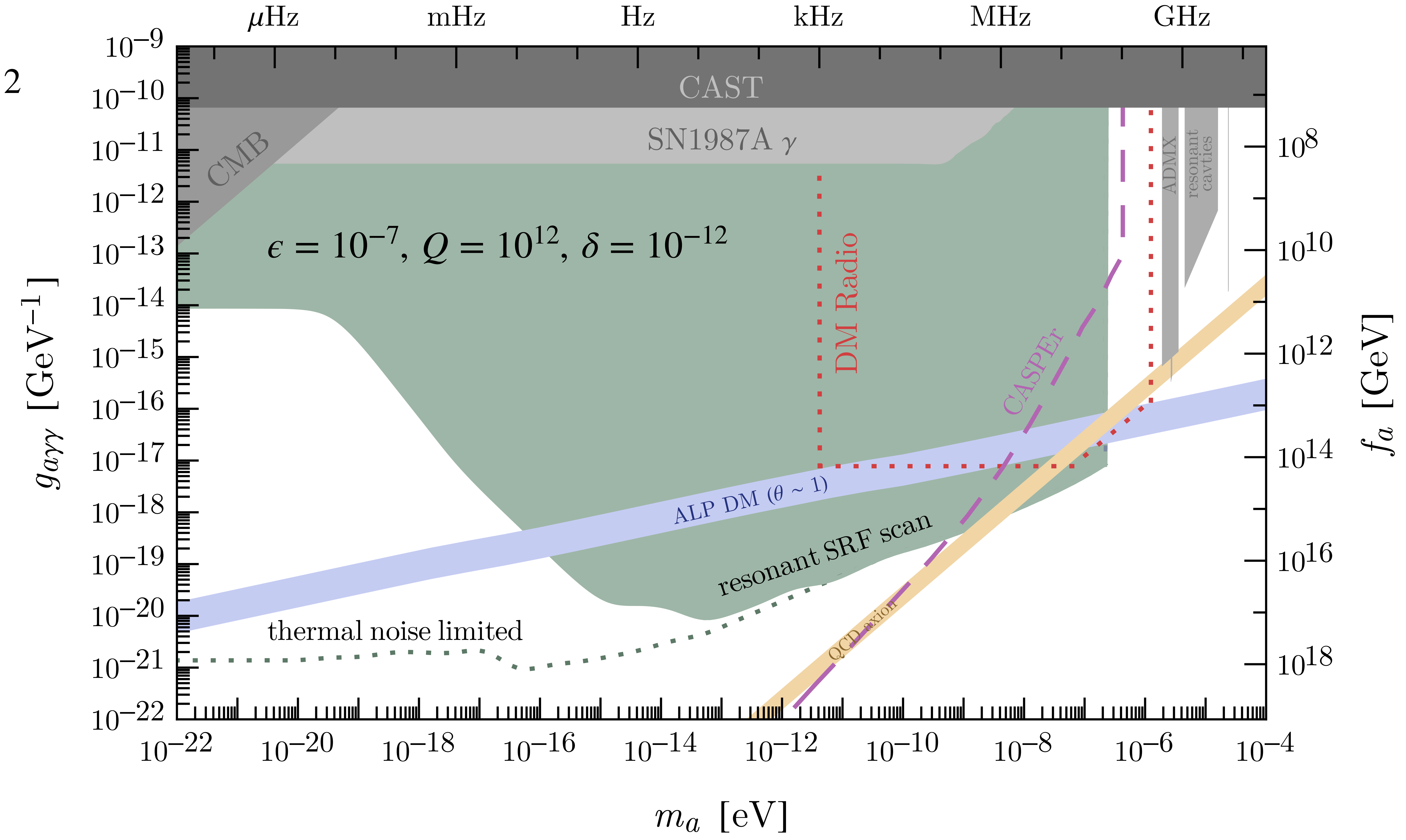
If $V = V_{\mathrm{LC}} = 1\,\mathrm{m}^3$ and thermal noise limited:

$$\frac{\mathrm{SNR}}{\mathrm{SNR}_{\mathrm{LC}}} \sim \frac{\omega_1}{m_a}\left(\frac{Q}{Q_{\mathrm{LC}}}\right)^{1/2}\left(\frac{T_{\mathrm{LC}}}{T}\right)^{1/2}\left(\frac{B}{B_{\mathrm{LC}}}\right)^2$$

Probes QCD axion in mass range complementary to other proposals

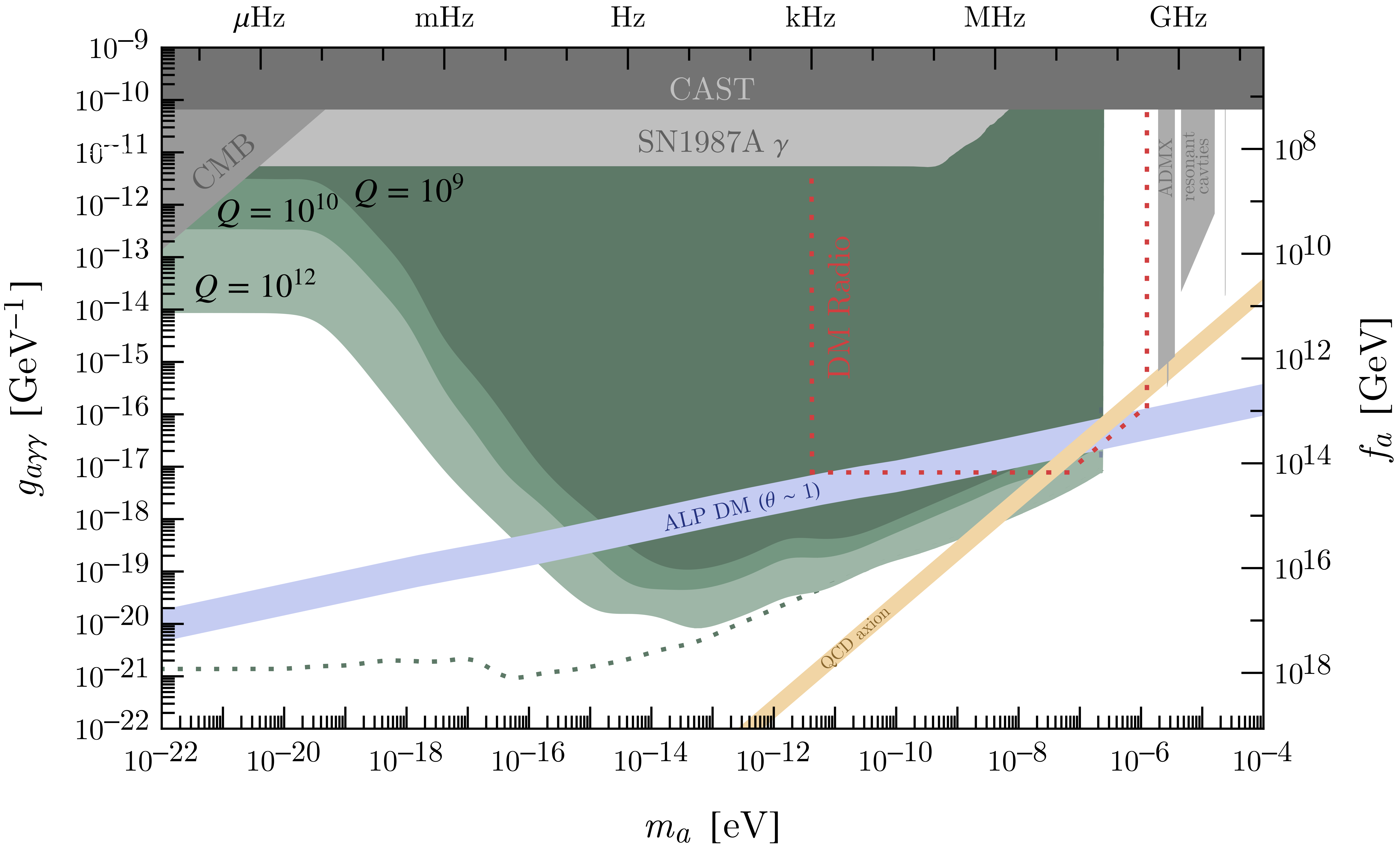
Sensitivity survives as $m_a \to 0$, probes ALP DM and fuzzy DM

Most parameter space for low m_a can be covered broadband, without need to scan



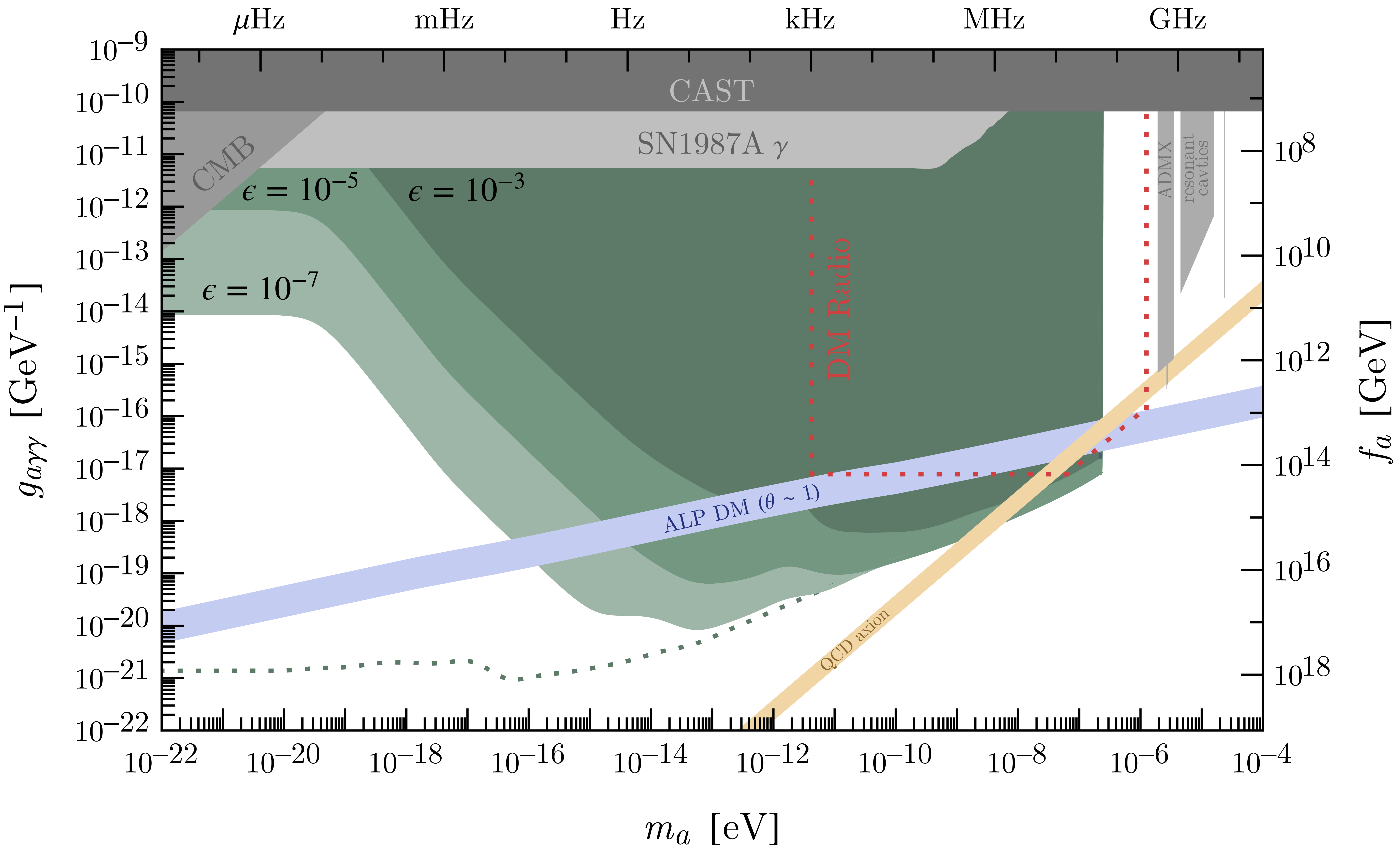
2007.15656: A. Berlin, R. Tito D’Agnolo, S. Ellis, K. Zhou

Robustness of Reach



Higher Q reduces all noise sources, $Q \sim 2 \times 10^{11}$ already achieved in SRF cavities

Robustness of Reach



Lower ϵ increases leakage noise, but reach almost unchanged at high m_a

Outlook

Wide ranging potential reach, including:

- QCD axion in unexplored region
- Entire viable mass range of fuzzy DM
- 10 orders of magnitude in m_a for ALP DM

Multidisciplinary experimental programme required!

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Multidisciplinary experimental programme required!

- Cavity design for high B and Q, wide scan
- Active feedback systems for ϵ and δ
- Overcoupled quantum limited readout